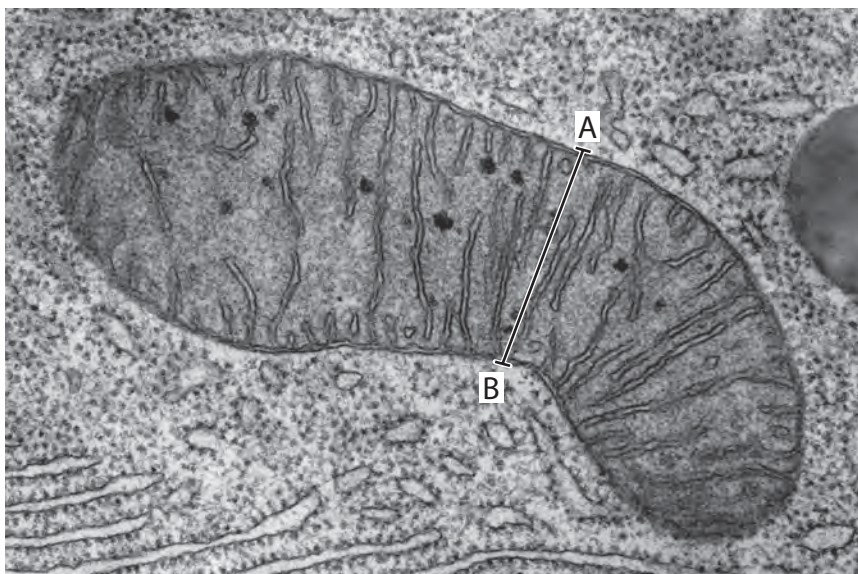


7 The photograph shows a mitochondrion.



(Source: © Science History Images/Alamy Stock Photo)

(a) The magnification of this photograph is $\times 23\,500$.

(i) Name the type of microscope that would be used to produce this image.

(1)

(ii) Calculate the actual width of this mitochondrion between points A and B.

Give your answer in μm to **three** significant figures.

(2)

Answer μm

(b) Respiration in a cell is composed of several processes.

(i) Which describes how ATP is synthesised by chemiosmosis?

(1)

- ☐ **A** hydrogen ions are pumped from the matrix to the intermembrane space through ATP synthase
- ☐ **B** hydrogen ions diffuse from the intermembrane space to the matrix through ATP synthase
- ☐ **C** hydrogen ions diffuse from the intermembrane space to the stroma through ATP synthase
- ☐ **D** hydrogen ions jump from the inner membrane to the outer membrane and out through ATP synthase

(ii) Amytal is a drug that inhibits the electron transport chain in respiration.

Which row of the table shows the changes caused by amytal when it is present in a cell?

(1)

	Oxygen used	Krebs cycle activity
☐ A	decreases	increases
☐ B	decreases	decreases
☐ C	increases	increases
☐ D	increases	decreases

(iii) Which molecule is required in the conversion of pyruvate to a 2 carbon compound in the link reaction of respiration?

(1)

- ☐ **A** oxidised NAD
- ☐ **B** oxidised NADP
- ☐ **C** reduced NAD
- ☐ **D** reduced NADP

(c) Name **three** products of Krebs cycle.

(2)

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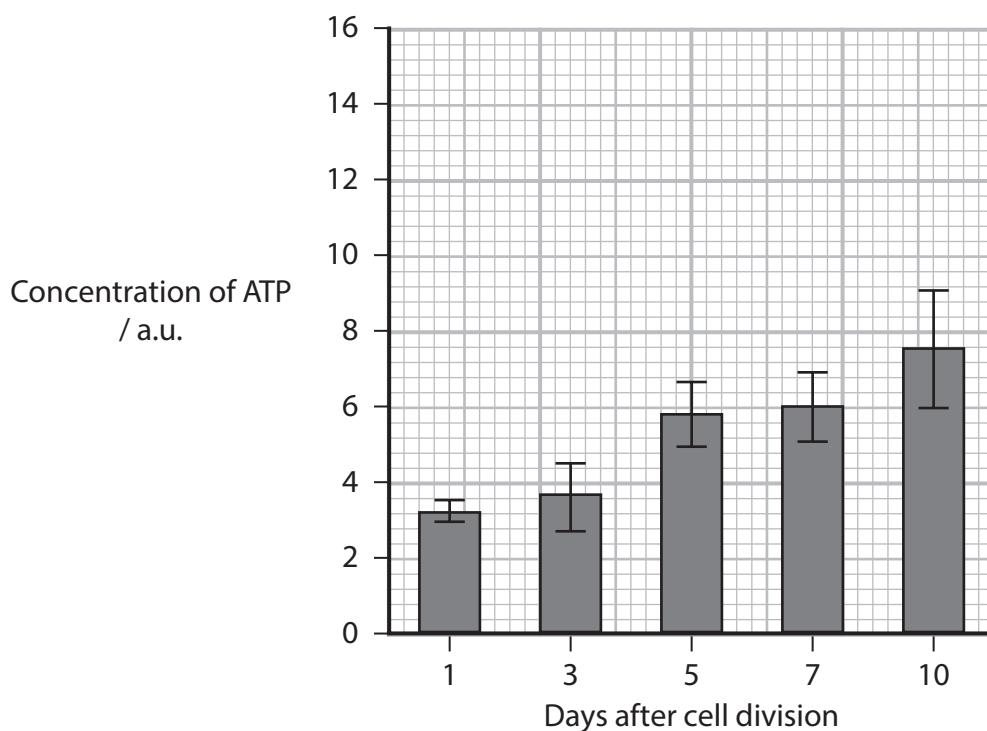
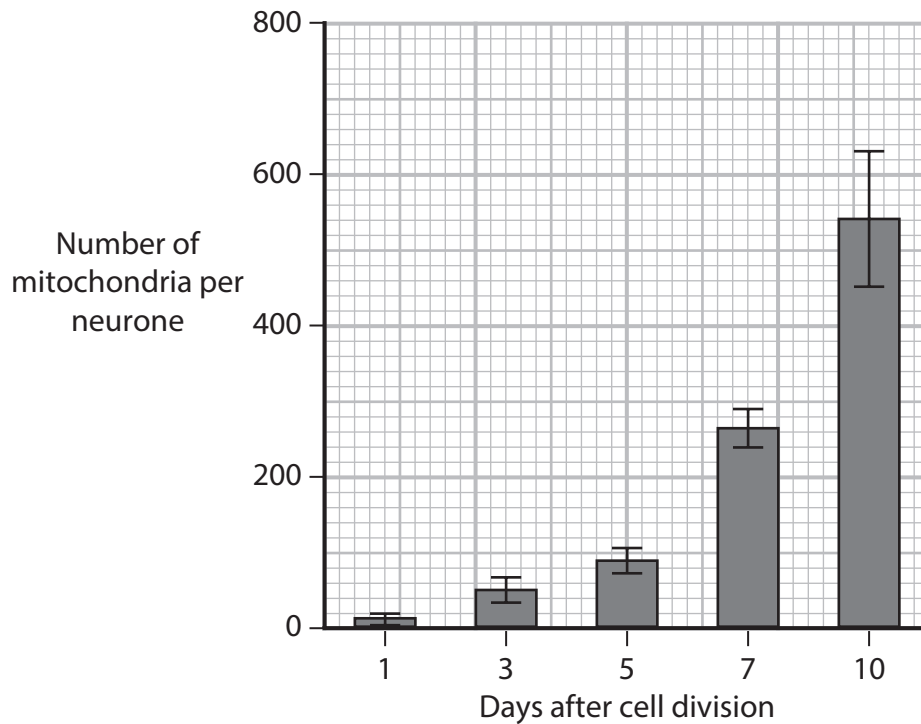
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- (d) As the brain develops some stem cells divide and differentiate into mature neurones, which requires ATP.

Mature neurones develop axons and form synapses with adjacent cells.

In an investigation, the number of mitochondria and the quantity of ATP made were measured after cell division and during differentiation of brain cells.

The graphs show the results of this investigation.



(Source: <https://pubmed.ncbi.nlm.nih.gov/23212379/>)

Comment on the conclusions that can be made from this investigation.

(4)

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(Total for Question 7 = 12 marks)
